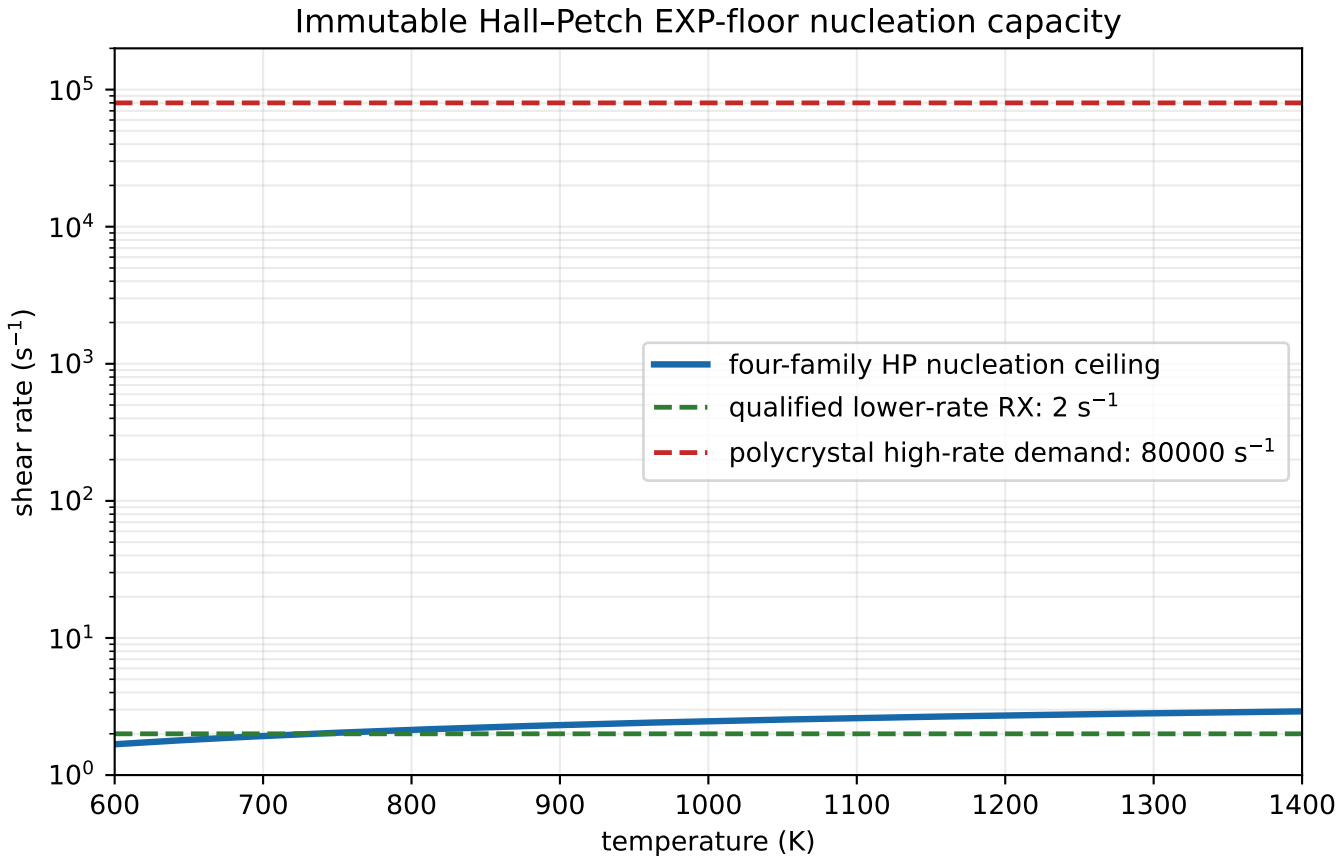
Immutable Hall–Petch EXP-floor nucleation capacity
shear rate (s^{-1})
temperature (K)
four-family HP nucleation ceiling
qualified lower-rate RX: 2 s^{-1}
polycrystal high-rate demand: 80000 s^{-1}
10^0
10^1
10^2
10^3
10^4
10^5
600
700
800
900
1000
1100
1200
1300
1400